Python Programming Project

Project Title: AI Resume Analyzer & Job Matching System

Course: Fundamentals of AI and ML

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INTRODUCTION

This project is designed to analyze a candidate's resume and compare it with a given job description using basic Artificial Intelligence techniques. Instead of manually checking resumes, the system calculates a similarity score, identifies missing skills, and provides improvement suggestions.

The system uses Natural Language Processing (NLP) concepts like TF-IDF and cosine similarity to evaluate how closely a resume matches job requirements. It also generates an ideal resume example to guide users in improving their profile.

PROBLEM STATEMENT

Candidates often apply for jobs without knowing whether their resume matches the job requirements. Recruiters also spend significant time filtering resumes manually.

Most systems do not provide detailed feedback or highlight missing skills. Therefore, there is a need for a system that can:

- Evaluate resume relevance
- Identify skill gaps
- Provide actionable suggestions

FUNCTIONAL REQUIREMENTS

- Input user details (Name, Skills, Projects, Soft Skills)
- Load job description
- Analyze resume text
- Calculate similarity score (%)
- Extract detected skills
- Identify missing skills
- Generate feedback
- Generate ideal resume sample

NON-FUNCTIONAL REQUIREMENTS

- User-friendly console interface
- Fast processing time
- Accurate text comparison
- Scalable for future improvements
- Clear and understandable output

SYSTEM ARCHITECTURE

Input Layer → Processing Layer → Output Layer

Inputs: Resume data (skills, projects, soft skills), job description

Processing: TF-IDF vectorization, cosine similarity, skill extraction

Output: Resume score, detected skills, missing skills, feedback, ideal resume

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DESIGN DIAGRAMS

Use Case Diagram: User → Resume Analyzer → Output Results

Workflow Diagram: Start → Input Resume → Analyze → Calculate Score → Generate Feedback → Display → End

Sequence Diagram: User → Input Module → Analyzer → Scoring Module → Feedback Module → Output

Component Diagram: Input Module, NLP Processing Module, Scoring Module, Feedback Generator

ER Diagram: Not applicable (no database used)

DESIGN DECISIONS & RATIONALE

• Used Python due to simplicity and strong library support
• Chose TF-IDF for efficient text representation
• Used cosine similarity for accurate comparison
• Predefined skill list for simplified extraction
• Console-based interface to focus on core logic
• Added ideal resume generation for better user guidance

IMPLEMENTATION DETAILS

• Implemented using Python functions and modular structure
• Used TfidfVectorizer from sklearn for text processing
• Used cosine_similarity to compare resume and job description
• Extracted skills using keyword matching
• Generated score based on similarity percentage
• Provided feedback based on score and missing skills

SCREENSHOTS / RESULTS

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🤖 AI Resume Analyzer (Demo Version)

Enter your details:

Name: pranjal wranjal
Skills (comma separated): python,html,java
Projects: vityarthi
Soft Skills: communication

📊 RESULTS
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Resume Score: 24.68%
Detected Skills: python, html, communication

🧠 FEEDBACK
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⚠️ Resume needs improvement.

❗ Missing Skills:
- excel
- machine learning
- data analysis
- problem-solving
- sql

💡 Suggestions:
- Add missing technical skills by including tools and technologies mentioned in the job description.
- Improve project descriptions by clearly explaining what you built, which technologies you used, and the impact or results of your work.
- Use strong action verbs like 'developed', 'designed', 'implemented', or 'analyzed' to make your experience sound more impactful.
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🌟 IDEAL RESUME (100% MATCH EXAMPLE)
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Name: pranjal wranjal

Skills:
python, sql, data analysis, machine learning, excel, communication, problem-solving

Projects:
- Developed a machine learning model using Python to analyze large datasets, improving prediction accuracy by 20%.
- Designed and implemented a data analysis dashboard using Python, SQL, and Excel to visualize key insights and support decision-making.

Soft Skills:
Communication, Problem-solving

Summary:
A results-driven candidate with strong technical expertise in Python, SQL, machine learning, and data analysis.
Experienced in building impactful projects and delivering data-driven solutions aligned with industry requirements.
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Example Output:

- Resume Score: 24.68%
- Detected Skills: Python, HTML, Communication
- Missing Skills: Machine Learning, Excel, etc.
- Feedback with suggestions

TESTING APPROACH

- Tested with different skill combinations
- Checked low, medium, and high similarity cases
- Verified missing skill detection
- Tested edge cases (empty input, minimal data)
- Validated feedback accuracy

CHALLENGES FACED

- Understanding TF-IDF and similarity calculation
- Extracting meaningful skills from text
- Designing realistic feedback logic
- Handling variations in user input
- Maintaining simplicity while keeping functionality useful

LEARNINGS & KEY TAKEAWAYS

- Learned basics of Natural Language Processing
- Understood TF-IDF and cosine similarity
- Improved Python programming and modular coding
- Gained knowledge of resume screening systems
- Learned how AI can solve real-world problems

FUTURE ENHANCEMENTS

- Add GUI using Tkinter or web interface
- Use real job datasets
- Integrate with job portals
- Use advanced NLP models (like BERT)
- Add resume upload (PDF/DOC support)

REFERENCES

- Python Documentation
- Scikit-learn Documentation
- NLP tutorials and online resources